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A01: Colab and other Tools Practice Run
ITAI 1378 – Computer Vision
Professor Patricia McManus

GitHub Repository:
https://github.com/Nichelleg/jupyter-exploration

What I Did:

For this lab, I worked with GitHub, GitHub Desktop, Google Colab, and a Jupyter Notebook. I already had a GitHub account with the username **Nichelleg** from working on assignments in previous classes in the Artificial Intelligence program at Houston City College. I also already had GitHub Desktop installed on my laptop, so I did not need to create a new account or install the desktop application.

I also had an existing Google Colab account, which is the environment I chose to use for this assignment. Although I have Visual Studio Code installed on my laptop and have used it for other coursework, Colab was the simplest option for completing the required Jupyter Notebook exercises. In Colab, I created a notebook that included both a Markdown cell and a Python code cell. I entered the sample Markdown text and ran a simple Python print() statement. I then downloaded the notebook as an .ipynb file so it could be added to my GitHub repository, committed, and pushed to GitHub.

For my coursework, I generally store assignment files locally under **C:\Source_Code** on my laptop. I prefer keeping a local copy because, in the past, I have lost files that were stored in the HCC cloud environment. Maintaining my own organized local folders gives me more confidence that I know exactly where my work is stored and makes it easier to manage files across multiple classes.

There was some initial confusion when I created the repository for this lab. I did not remember that I had already completed a similar Jupyter and GitHub assignment for my NLP class in a previous semester. Because of that, I had already created a repository and notebook with very similar names. At first, I could not understand why GitHub Desktop appeared to be connecting to an unexpected folder or why files seemed to be showing up in the wrong location.

After looking more closely at the repositories and file paths, I realized that I was dealing with remnants from the older NLP assignment as well as the new Computer Vision assignment. I renamed and reorganized the older files so I could distinguish the NLP work from the current Computer Vision work. I eventually established the correct local repository at **C:\Source_Code\jupyter-exploration** and verified that the correct notebook, README file, and Git repository information were present.

Once the files were organized correctly, I was able to commit the notebook and README changes and push them to the public GitHub repository. The assignment instructions require the repository to contain the notebook and be accessible through a public GitHub link, which I verified before moving on to the written summary.

What I Learned:

Although I had used GitHub, GitHub Desktop, Google Colab, Jupyter Notebooks, and VS Code in previous classes, this lab reminded me that familiarity with a tool does not always mean that every step remains easy to recall. I did not take summer classes, and after being away from these tools for several months, I realized that I had forgotten more of the workflow than I expected.

Even simple operations such as identifying the correct local repository, understanding which folder GitHub Desktop was tracking, and remembering how the local files related to the online repository were initially confusing. I knew the general concepts, but some of the specific steps were no longer automatic.

This lab also helped reinforce the relationship between a local repository and a remote GitHub repository. A file placed in a local repository is not automatically updated on GitHub. The changes must first be recognized by Git, committed locally, and then pushed to the remote repository. GitHub Desktop provides a visual way to manage this process, but it still depends on the correct local directory being connected to the correct online repository.

Another lesson from the lab was the importance of organized naming conventions. Reusing generic names such as jupyter-exploration or My_First_Notebook.ipynb can become confusing when the same type of introductory assignment is completed in multiple courses. Because I had already completed a similar exercise for NLP, the repeated filenames and repository names created unnecessary confusion. Going forward, I plan to be more deliberate about identifying older course files so I can quickly distinguish work from different semesters and classes.

The lab also gave me a useful reminder about skill retention. I have worked with these tools before, but I had not used some of them regularly during the summer. The fact that relatively simple tasks were stumping me at first showed me how quickly procedural knowledge can become rusty when it is not used consistently. Repeating the process was useful because it refreshed skills that I will need throughout the Computer Vision course.

When I became confused about the repository naming and file paths, I consulted ChatGPT for help troubleshooting the situation. I used it to help identify why GitHub Desktop was referencing an older repository, distinguish between the old NLP files and the new Computer Vision files, and work through the naming and folder structure. I completed the actual file management, GitHub changes, commits, and repository updates myself.

Comments:

One thing I took away from this lab is that introductory exercises can still be useful even for students who have previously worked with the same tools. In my case, the lab was less about

seeing GitHub or Jupyter for the first time and more about refreshing workflows that I had not used recently.

The experience also reinforced why file organization matters as the number of classes, repositories, notebooks, and projects increases. Keeping clear local folders and identifying older projects by course will make it easier to manage future assignments and avoid mixing work from different semesters.

AI Use Disclosure: ChatGPT was used as a troubleshooting and learning aid during this assignment when I encountered confusion involving GitHub Desktop, local repository paths, and similarly named repositories from a previous course. I used the guidance to understand and resolve the file organization issue, while completing the actual GitHub, Colab, and file management steps myself.